

TURTLE SHELL MATERIAL



TEAM 03

Name of material and the natural properties

Made of bone covered by ourter layer

- Compressive strength
- Hardness
- Heavy
- Tensile strenght



How the material was developed

- Strong and hard
- Studying the structure of a turtle shell
- Inspiration and designed a layered engineering material
- Biological research
- Scanning



Key properties and benefits

- High strength and hard
- Lightweight
- Long durability
- Impact resistance
- Good protection and safety



HOW HARD IS A TURTLE SHELL?

How much pressure can a turtle shell withstand?



Its intended application

- Motorcycle helmets
- Body armor
- Sports protective gear



Why it is suitable

- Its lightweight to wear
- Absorb and distribute impact forces



One possible future use

- Space suits
- Spacecraft outer panels

Future research

- Enviromental archives
- Genetic pathway
- Dental and medical research



One limitation

- Banned worldwide to kill turtles
- Shell is not flat or smooth
- Shell pieces are very small
- Heat damages
- Trying to shape can dry out and crack



Invested area

- Military
- Sports
- Transport industries



THE

END

Team 03

Theekshana

Malith

Pehan

Rusinu

Thewmini